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Editors
Imaging Neuroscience

Dear Editors,

Please consider the Research Article "Contralateral-Normalized Deformation and Stationary Log-Velocity Decomposition after Chronic Stroke: Geometric Characterization and Aphasia Prediction in the Aphasia Recovery Cohort" for publication in *Imaging Neuroscience*.

This study addresses a recurring interpretive problem in lesion neuroimaging: registration fields near chronic lesions can be scientifically useful without being physical measurements of tissue displacement or pressure. We introduce a contralateral-normalized, lesional-only deformation proxy, construct a positive-Jacobian stationary log-velocity embedding, and apply an explicitly bounded Helmholtz–Hodge decomposition. Every log-domain case is audited by exponentiating the estimated velocity and comparing the reconstruction with its input deformation.

The main scientific result is deliberately negative and useful. In 210 participants from the openly available Aphasia Recovery Cohort, deformation features produced only a small and uncertain reduction in held-out Western Aphasia Battery Aphasia Quotient error. Hodge descriptors were descriptively associated with aphasia severity but did not improve nested cross-validated prediction beyond conventional lesion or displacement features. The manuscript distinguishes this geometric analysis from physical pressure inference and reports the numerical, registration-sensitivity, and statistical checks needed to support that boundary.

The work fits *Imaging Neuroscience* because it combines open human neuroimaging, deformable registration, reproducible computational methodology, and out-of-sample brain–behavior modeling. Code, tests, aggregate results, and LaTeX sources are publicly available in the ARCDeformationAnalysis repository; the source dataset is available from OpenNeuro.

OpenAI Codex assisted with code refactoring, statistical implementation, reproducibility checks, and language editing. The author reviewed the analysis and accepts responsibility for the submitted code and text. The corresponding disclosure appears in the manuscript.

Before submission, the author will confirm originality and exclusive consideration and supply the final funding and competing-interest declarations in the manuscript and submission system.

Sincerely,

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